

CONFERENCE TALK - 15 MINUTES

Reconstructing Hidden Common Driver Dynamics

Can shared latent dynamics be recovered from two observed driven systems?

A results-focused tour through Mapper-Coach, AniSOM/ASOM, and the benchmark experiments in this repository.

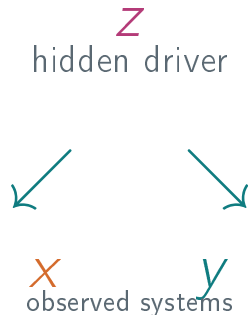
Mapper-Coach: [arXiv:2105.02322](https://arxiv.org/abs/2105.02322) ASOM/AniSOM: [arXiv:2504.01811](https://arxiv.org/abs/2504.01811)

Problem Setup

Two observed systems contain their own dynamics plus a shared imprint from an unobserved driver.

- ▶ Observed time series: x and y .
- ▶ Hidden process: z , a common driver.
- ▶ Time-delay embedding reconstructs state spaces from histories.
- ▶ The target is the latent trajectory, up to an admissible transformation.

Fork structure



Both cited papers exploit the redundant imprint of a shared latent driver in driven observations.

Method Landscape

- ▶ **MaCo**: Mapper-Coach neural bottleneck trained through prediction.
- ▶ **ASOM/AniSOM**: anisotropic self-organizing map for shared/self dynamics.
- ▶ **ShRec**: recurrence-structure shared reconstruction baseline.
- ▶ **DCCA**: deep canonical correlation analysis.
- ▶ **DCA**: dynamical component analysis.
- ▶ **CCA**: linear correlated-components baseline.
- ▶ **PCA / ICA / kPCA**: classical extraction and source separation.
- ▶ **SFA / Random**: slow-feature method and sanity-check control.

Comparison question

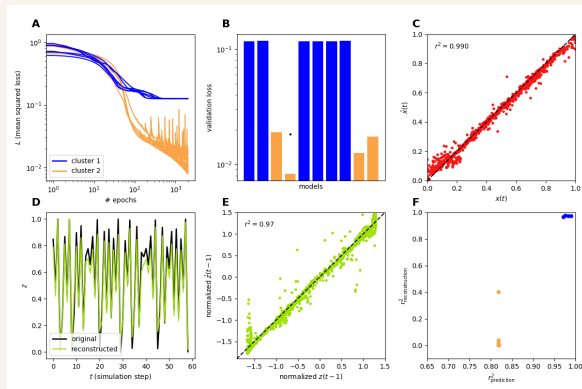
Which kind of structure best exposes the hidden driver: predictive bottlenecks, attractor geometry, correlated features, or generic variance/source decomposition?

Mapper-Coach Intuition

MaCo learns to predict one driven system using its own past and information extracted from the other driven system.

- ▶ Mapper compresses the reconstructed state of one observation.
- ▶ Coach uses that bottleneck to predict the other observation.
- ▶ Successful prediction makes the bottleneck align with the hidden driver.

Example logistic-map run: high prediction quality and high latent reconstruction quality appear together.



ASOM / AniSOM Intuition

The ASOM paper frames hidden-driver inference through time-delay embedding, intrinsic dimension, mutual dimension, and anisotropic SOM training.

- ▶ The observed attractor is treated as a structured manifold.
- ▶ Anisotropic training emphasizes directions associated with common dynamics.
- ▶ The output gives a geometry-first unsupervised driver estimate.

Conceptual split

Observed manifold: **self + shared**

Mutual structure: **shared driver**

AniSOM output: **driver estimate**

This complements MaCo: one method learns through prediction, the other through attractor geometry.

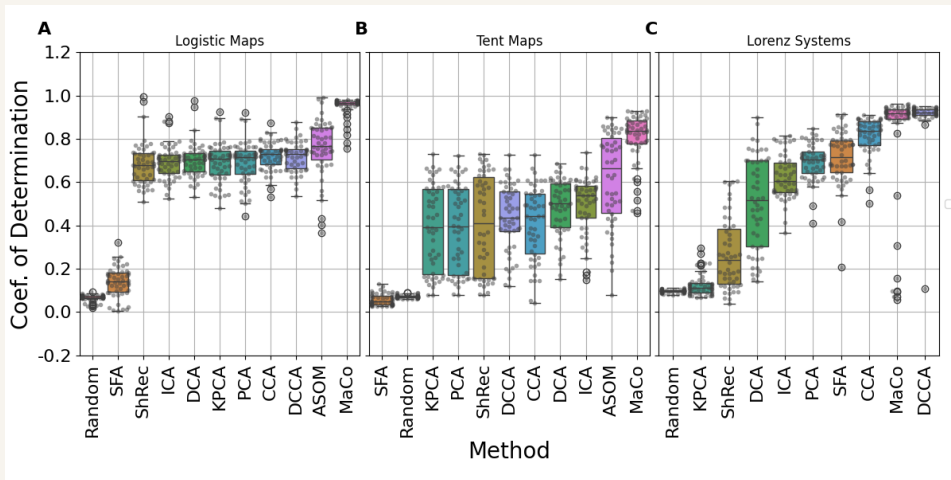
Experiment Design

- ▶ **Logistic maps:** discrete chaotic maps with known hidden common input.
- ▶ **Tent maps:** nonlinear discrete-map family with sharper behavior.
- ▶ **Lorenz systems:** continuous-flow setting with harder reconstruction geometry.
- ▶ **Scale:** 50 realizations per main comparison family.

Figures and final CSVs are tracked under `paper_artifacts`.

Family	Metric	Source
Logistic	Corr. quality	logmaps
Tent	Corr. quality	tentmaps
Lorenz	Corr. quality	lorenz

Main Result Across Datasets



leads on logistic and tent maps. DCCA leads on Lorenz, with MaCo still competitive.

Result Numbers

Dataset	Top result	Strong comparators
Logistic maps	MaCo 0.9491	ASOM 0.7571
Tent maps	MaCo 0.8011	ASOM 0.6168
Lorenz	DCCA 0.9063	MaCo 0.8179; CCA 0.8124

The headline is not that one method wins everywhere. Hidden-driver reconstruction depends strongly on the geometry of the observed system.

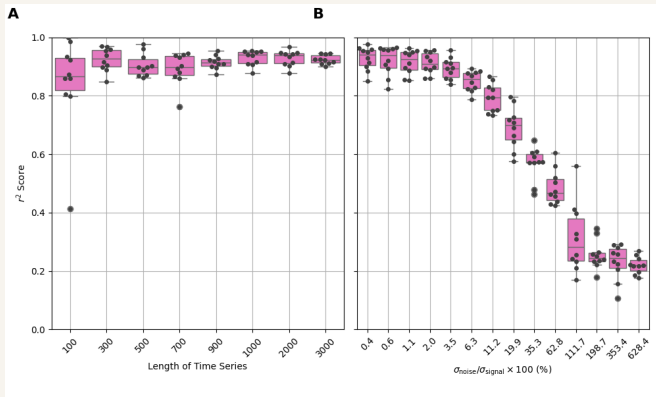
- ▶ MaCo dominates the discrete-map benchmarks.
- ▶ ASOM is the strongest map-family manifold baseline.
- ▶ Lorenz elevates DCCA and narrows the MaCo/CCA gap.

Means computed from the final CSV result tables.

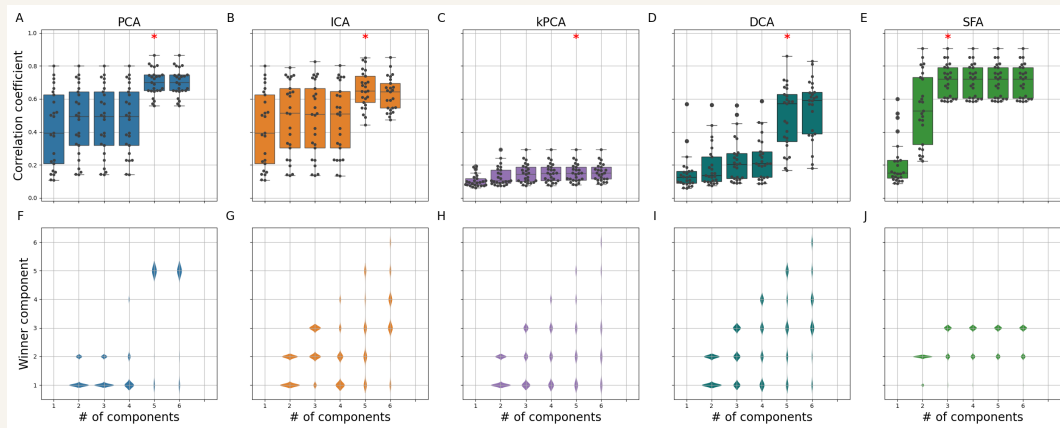
Robustness

MaCo remains useful with moderate data and low-to-mid noise.

- ▶ Strong performance already at length 100.
- ▶ Stabilizes around length 300 and beyond.
- ▶ Noise produces smooth degradation rather than abrupt failure.



Hyperparameter Context



Lorenz tuning context for PCA, ICA, kPCA, DCA, and SFA. Baseline strength depends on component and winner choices.

Interpretation

Prediction can reveal latent structure	Geometry remains competitive	Continuous flows are harder
MaCo supports the Mapper-Coach claim: a minimal predictive bottleneck can align with the common hidden driver.	ASOM/AniSOM offers a strong unsupervised manifold baseline, especially on discrete chaotic maps.	Lorenz results show that method ranking changes when dynamics and embeddings become more complex.

Together, the papers motivate both neural and geometric approaches to hidden-driver inference.

Takeaways and References

- ▶ Hidden common drivers are recoverable when their imprint is redundantly encoded.
- ▶ MaCo is strongest on the discrete-map benchmarks in this repo.
- ▶ Lorenz and noise analyses show where reconstruction becomes more demanding.

Benko and Somogyvari. *Reconstructing shared dynamics with a deep neural network*. arXiv:2105.02322.

<https://doi.org/10.48550/arXiv.2105.02322>

Benko, Stippinger, Bencze, Bazso, Telcs, and Somogyvari. *Inference of hidden common driver dynamics by anisotropic self-organizing neural networks*. *Neural Networks* 194, 108113.

<https://doi.org/10.1016/j.neunet.2025.108113>

Repository artifacts: [paper_artifacts/figures/](#) and [paper_artifacts/results/final/](#).